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ORIGINAL ARTICLE

Preverbal visual assessment for screening cerebral visual impairment: Diagnostic accuracy and clinical utility

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Abstract

Aim: To evaluate the relationship between Preverbal Visual Assessment (PreViAs) results and cerebral visual impairment (CVI) diagnosis.

Method: This single-center retrospective chart review included children who completed a CVI interdisciplinary clinic or occupational therapy vision evaluation between May 2018 and May 2023 and had a completed PreViAs. Data were extracted from the electronic medical record. Data analysis included sensitivity, specificity, area under the curve, and Fisher's exact test.

Results: Of the 205 charts screened, 98 met study criteria (81 with CVI, 17 without); ages ranged from 3 to 50 months. The mean age of children with CVI was 15.41 months (SD = 8.93) and without was 13.24 months (SD = 5.98). Among the PreViAs domains, visual processing showed the highest frequency of concern (CVI = 94%; non-CVI = 76%), while visual attention had the lowest (CVI = 72%; non-CVI = 41%). Visual processing demonstrated the highest sensitivity (93.8%), whereas visual attention showed the best balance of sensitivity (71.6%) and specificity (58.8%). Area under the curve ranged from 0.59 to 0.66. All domains demonstrated significant differences between children with and without CVI (Fisher's exact test, $p < 0.05$).

Interpretation: Effective screening tools are critical for detection of CVI. The PreViAs is a sensitive screening tool in children at high risk for CVI. Children with concerns on the PreViAs should be referred for an interdisciplinary CVI evaluation.

Cerebral visual impairment (CVI) is the most common cause of bilateral visual impairment in children in high-income countries.[1] CVI is a neurological condition caused by damage or malformation to the areas of the brain responsible for processing visual information before, during, or shortly after birth.[2,3] The most common etiologies of CVI include periventricular leukomalacia, congenital brain malformations, hypoxic–ischemic encephalopathy, intraventricular hemorrhage, and preterm birth.[4–6] CVI can also result from genetic or metabolic disorders, central nervous system infections, and traumatic brain injuries. CVI commonly co-occurs with other neurological conditions and developmental disorders, such as cerebral palsy, with as many as 60% to 70% of children with cerebral palsy also having CVI.[7] Additionally, seizures are a prevalent co-occurring disorder, affecting up to 50% of children with CVI.[8] This strong evidence linking common conditions to an increased risk of CVI makes a compelling argument for proactive CVI screening to enable early intervention and educational accommodations to improve developmental outcomes.[9]

The presence of CVI can negatively impact development, including cognition, language, and motor skills.[9,10] CVI is a heterogeneous condition with varying levels of severity and symptoms depending on the degree of neurological

Abbreviations: AOC, area of concern; CVI, cerebral visual impairment; MON, continue to monitor; PreViAs, Preverbal Visual Assessment.

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impairment and the presence of comorbid developmental conditions. Children with CVI may experience difficulties recognizing colors, shapes, and reading, leading to delayed learning.[11,12] They may struggle with visual attention, tracking moving objects, and processing visual stimuli, particularly in crowded or unfamiliar settings.[9,13] This can make certain environments, like classrooms, feel visually overwhelming. Children with CVI learn new information differently from those without CVI. Typically developing children learn by observing their surroundings, interacting with people, and exploring visually, much of which happens through incidental learning, where knowledge is gained unintentionally during everyday activities and interactions. In contrast, children with CVI often depend on other senses, such as touch and hearing, to compensate for their impaired functional vision. Children with CVI require more targeted instruction and guidance for most aspects of learning, motor exploration, and social interactions.[14,15] Therefore, early screening and detection of CVI is crucial for timely, targeted interventions that maximize neuroplasticity and improve overall quality of life.[15]

Although the prevalence of CVI is increasing, CVI is often not diagnosed until preschool age or later.[16] Late diagnosis leads to missed opportunities for vision-specific, early intervention services.[17] A validated caregiver questionnaire is essential for effective CVI screening to enable earlier detection and diagnosis. CVI screening helps clinicians understand who should undergo a more comprehensive, interdisciplinary CVI evaluation.[18,9] The Preverbal Visual Assessment (PreViAs) is a caregiver screening questionnaire used to describe observations of visual behaviors of infants 24 months developmental age and under across four domains: visual attention, visual communication, visual motor coordination, and visual processing.[18] Using a sample of 220 infants, the PreViAs developed age-dependent cut-off scores to identify infants with delayed visual maturation. It has been validated in children aged 2 to 24 months.[19] To date, the PreViAs has not been tested as a screening tool for infants at high risk for CVI. Therefore, the study aimed to evaluate the relationship between PreViAs results and CVI diagnosis using retrospective data. The ethics board at our institution approved this study (IRB 2014–6879).

What this paper adds

- The Preverbal Visual Assessment (PreViAs) is a sensitive screening tool for cerebral visual impairment.
- Failed PreViAs should be defined as any domain scoring area of concern or continue to monitor, prompting referral for comprehensive evaluation.

METHOD

Study design

A retrospective study design was employed at a US pediatric academic medical center.

Participants

Participants at high-risk for CVI were included in the study if they (1) completed a diagnostic CVI evaluation between May 2018 and May 2023 in a CVI interdisciplinary clinic or an outpatient occupational therapy evaluation of functional vision, and (2) the caregiver completed a PreViAs questionnaire. Children were considered at high risk for CVI if they had a medical diagnosis (e.g. seizures, periventricular leukomalacia) or exhibited visual symptoms typically associated with CVI. Some children over 24 months were included in the study because clinicians determined, based on developmental age and preverbal status, that the PreViAs was the most appropriate screener.

Data abstraction

Charts were identified based on specific visit types used to schedule children in the CVI interdisciplinary clinic and/or for an occupational therapy functional vision evaluation. PreViAs data were extracted through a report run from an electronic medical record based on the CVI visit types. The data were verified by two reviewers (KH, RD). The first PreViAs was used if participants had more than one. Additionally, demographic data, including sex, age at the time of PreViAs completion, CVI diagnosis status, date CVI diagnosis was given, and the primary medical conditions associated with increased CVI risk, were collected. The interval between PreViAs administration and confirmation of CVI diagnosis was recorded. CVI diagnosis status and date were extracted from ophthalmology clinical notes. Formal and informal evaluations were administered by an ophthalmologist/optometrist, occupational therapist, teacher of students with visual impairment, and orientation and mobility specialist. CVI diagnosis was based on the following criteria: (1) medical history consistent with injury to the posterior visual pathway (developmental and birth history, neuroimaging findings); (2) CVI screening questionnaire (PreViAs) or visual history consistent with CVI; (3) identified visual function impairment (e.g. visual acuity, contrast sensitivity) not attributed to anterior visual pathway abnormalities (except for mild optic atrophy); and (4) presence of visual perception/functional vision deficit.

Measures

Preverbal Visual Assessment (PreViAs)

The PreViAs is a caregiver questionnaire developed to screen visual maturation in children up to 24 months

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developmental age.[18] It consists of 30 yes/no questions, divided into four domains: visual attention, visual communication, visual motor coordination, and visual processing. For scoring, 1 point is given for each visual behavior demonstrated, with a potential overall score of 30. The maximum scores for each domain are as follows: visual attention is 11, visual communication is 5, visual motor coordination is 13, and visual processing is 20. The PreViAs has validated age-dependent cut-off scores for each domain, categorizing each child into one of three concern categories: no concern, continue to monitor (MON), or area of concern (AOC).[19]

Statistical analysis

Descriptive statistics were used to describe the sample demographics and outcome measures. Categorical variables were summarized using frequencies and percentages, while interval variables were described using measures of central tendency and dispersion. CVI diagnosis was coded as yes/no; PreViAs domains were coded as positive = yes, negative = no.

Fisher's exact test with the Freeman–Halton extension was used to analyse the effect of each of the four PreViAs domain outcomes on CVI diagnosis. Odds ratios with 95% confidence intervals were reported. Area under the curve, sensitivity, and specificity analyses were calculated. Wilcoxon rank-sum tests were used to compare PreViAs outcomes between children with and without CVI within each age group (0–12 months, 13–24 months, and 25+ months). Data analysis was conducted using SAS statistical software, version 9.4 (SAS Institute, Cary, NC, USA). The two-sided significance level for all analyses was set at 0.05.

RESULTS

Of the 205 charts reviewed, 98 were included in this study. Charts were excluded if they had incomplete or missing PreViAs data (107, 52%). Demographic data for all included charts is in Table 1. Eighty-one children (83%) received a CVI diagnosis, with only 17 (17%) not diagnosed with CVI. The mean age of participants with CVI at the time of PreViAs completion was 15.41 months (SD = 8.93, range = 2–50 months)

TABLE 1 Demographics ($n = 98$).

Age at PreViAs completion (months)	**CVI diagnosis $n = 81$ Mean (SD)**		**No CVI diagnosis $n = 17$ Mean (SD)**	
Entire cohort (CVI = 81, no CVI = 17)	15.41 (8.93)		13.24 (5.98)	
0–12 months (CVI = 41, no CVI = 10)	8.59 (2.31)		9.00 (2.62)	
13–24 months (CVI = 28, no CVI = 6)	18.18 (2.65)		18.33 (2.73)	
25+ months (CVI = 12, no CVI = 1)	32.25 (6.34)		29 (0)	
Sex	***n* (%)**		***n* (%)**	
Male	50 (62)		11 (65)	
Female	31 (38)		6 (35)	
	CVI diagnosis $n = 81$		**No CVI diagnosis $n = 17$**	
Conditions at risk for CVI (examples of conditions within the category)[a]	***n* (%)**	**History of seizures *n* (%)**	***n* (%)**	**History of seizures *n* (%)**
Brain injury (IVH, hydrocephalus, TBI, stroke)	30 (37)	10 (33)	6 (35)	–
Genetic condition (BBSOAS, CHARGE syndrome, Kabuki syndrome, gene mutation)	22 (27)	7 (23)	4 (23.5)	2 (67)
Hypoxic/ischemic (HIE, anoxic brain injury, placental abruption)	11 (13.5)	5 (17)	–	–
Congenital structural abnormalities of the brain (brain malformation, schizencephaly, gray matter heterotopia)	6 (7)	2 (7)	2 (12)	–
Seizures—unexplained	3 (4)	3 (10)	1 (6)	1 (33)
Intrauterine drug exposure	3 (4)	–	1 (6)	–
Unknown	3 (4)	–	1 (6)	–
Central nervous system infection (congenital or acquired) (neonatal group B strep meningitis, congenital CMV, perinatal HSV encephalitis)	3 (4)	3 (10)	–	–
Anterior pathway disease	–	–	2 (12)	–
Total	81	30	17	3

Abbreviations: BBSOAS, Bosch-Boonstra-Schaaf Optic Atrophy syndrome; CMV, cytomegalovirus; CVI, cerebral visual impairment; HIE, hypoxic–ischemic encephalopathy; HSV, herpes simplex virus; IVH, intraventricular hemorrhage; PreViAs, Preverbal Visual Assessment; TBI, traumatic brain injury.

[a]Not a comprehensive list of conditions within categories.

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and 13.24 months (SD = 5.98, range = 4–25 months) without CVI. Additionally, participants were categorized into three groups based on age when the PreViAs was completed. The primary condition putting each participant at higher risk for CVI was identified (Table 1). These conditions were grouped into nine categories and reported based on CVI diagnosis status. The most frequent CVI high-risk condition was brain injury, and the second most frequent was genetic. A history of seizures was common in our sample. Children with seizures primarily associated with genetic conditions were categorized into the genetic category. Children with unexplained seizures were grouped into a separate category for seizures.

The PreViAs scores for participants with and without CVI are reported by domain in Table 2. The three levels of concern (AOC, MON, and no concern) are reported independently by domains, as well as with AOC and MON combined to represent an overall level of concern. Mean scores of the four PreViAs domains were also calculated by age group (0–12 months, 13–24 months, 25+ months) and are presented in Table 3. The mean differences were significant in the PreViAs between groups (with and without CVI) for all domains in participants aged 0 to 12 months. For those aged 13 to 24 months, only visual communication and visual attention were statistically different. None of the mean differences were significant in the 25+ months group.

To understand the level of concern (AOC alone or AOC + MON) that should prompt further evaluation, sensitivity and specificity of the PreViAs were evaluated using two analyses (Tables 4 and 5). When concern was defined

TABLE 2 Frequency of concern on PreViAs by diagnosis status.

		Visual communication *n* (%)	Visual motor coordination *n* (%)	Visual processing *n* (%)	Visual attention *n* (%)
CVI *n* = 81	AOC	41 (51)	42 (52)	45 (56)	32 (40)
	MON	27 (33)	30 (37)	31 (38)	26 (32)
	NONE	13 (16)	9 (11)	5 (6)	23 (28)
No CVI *n* = 17	AOC	8 (47)	6 (35)	7 (41)	3 (18)
	MON	2 (12)	5 (29)	6 (35)	4 (23.5)
	NONE	7 (41)	6 (35)	4 (23.5)	10 (59)
CVI *n* = 81	AOC + MON	68 (84)	72 (89)	76 (94)	58 (72)
	NONE	13 (16)	9 (11)	5 (6)	23 (28)
No CVI *n* = 17	AOC + MON	10 (59)	11 (65)	13 (76)	7 (41)
	NONE	7 (41)	6 (35)	4 (24)	10 (59)

Abbreviations: AOC, area of concern; CVI, cerebral visual impairment; MON, area to monitor; NONE, no concern; PreViAs, Preverbal Visual Assessment.

TABLE 3 Mean (SD) PreViAs scores by age group and CVI status, with Wilcoxon rank-sum test results and effect sizes (r).

Age group	Diagnosis	Visual communication Mean (SD)	Visual motor coordination Mean (SD)	Visual processing Mean (SD)	Visual attention Mean (SD)
0–12 months	CVI (*n* = 41)	1.8 (1.5)	2.9 (2.19)	3.2 (2.36)	6.1 (2.89)
	No CVI (*n* = 0)	3.0 (1.8)	5.4 (3.05)	5.9 (4.65)	8.9 (2.98)
	p (r)	$p = 0.042$[a] (r = 0.28)	$p = 0.011$[a] (r = 0.37)	$p = 0.025$[a] (r = 0.32)	$p = 0.007$[a] (r = 0.40)
13–24 months	CVI (*n* = 28)	2.7 (1.53)	5.5 (2.72)	6.4 (3.4)	8.4 (2.22)
	No CVI (*n* = 6)	3.8 (1.47)	7.7 (3.14)	9.5 (5.01)	9.7 (2.42)
	p (r)	$p = 0.048$[a] (r = 0.25)	$p = 0.090$ (r = 0.20)	$p = 0.113$ (r = 0.18)	$p = 0.048$[a] (r = 0.25)
25+ months	CVI (*n* = 12)	3.0 (1.91)	7.2 (4.17)	9.7 (5.55)	8.4 (2.64)
	No CVI (*n* = 1)	4.0 (0)	7.0 (0)	9.0 (0)	11.0 (0)
	p (r)	$p = 0.890$ (r = 0.04)	$p = 0.784$ (r = 0.08)	$p = 1.000$ (r = 0.00)	$p = 0.334$ (r = 0.27)

Note: Effect size (r) interpretation: small = 0.10, medium = 0.30, large = 0.50.

The *p*-values for the 25+ months group should be interpreted with caution. The extremely small sample size (*n* = 1) in the no CVI group limits the statistical power and increases the risk of small sample bias.

Abbreviations: AOC, area of concern; CVI, cerebral visual impairment; MON, area to monitor; NONE, no concern; PreViAs, Preverbal Visual Assessment.

[a]Statistical significance at $p < 0.05$.

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TABLE 4 Sensitivity, specificity, AUC, and OR (95% CI) by visual domain when concern is defined as AOC only (no concern + MON vs AOC) for all study participants ($n = 98$; CVI-positive = 81, CVI-negative = 17).

Domain	Sensitivity (95% CI)	Specificity (95% CI)	AUC (95% CI)	OR (95% CI)	p[a]
Visual communication	50.6% (39.8, 61.4)	52.9% (31.0, 73.8)	0.52 (0.39, 0.65)	1.15 (0.40, 3.29)	1.00
Visual motor coordination	51.9% (41.0, 62.6)	64.7% (41.4, 82.7)	0.58 (0.46, 0.70)	1.97 (0.67, 5.85)	0.29
Visual processing	55.6% (44.7, 65.9)	58.8% (36.0, 78.4)	0.57 (0.45, 0.69)	1.79 (0.62, 5.16)	0.30
Visual attention	39.5% (29.6, 50.3)	82.4% (59.0, 93.8)	0.61 (0.49, 0.73)	3.05 (0.81, 11.46)	0.10

Abbreviations: AOC, area of concern; AUC, area under the curve; CI, confidence interval; CVI, cerebral visual impairment; MON, continue to monitor; OR, odds ratio.
[a]p-values from Fisher's exact test.

TABLE 5 Sensitivity, specificity, AUC, and OR (95% CI) by visual domain when concern is defined as AOC + MON (no concern vs AOC + MON) for all study participants ($n = 98$; CVI-positive = 81, CVI-negative = 17).

Domain	Sensitivity (95% CI)	Specificity (95% CI)	AUC (95% CI)	OR (95% CI)	p[a]
Visual communication	84.0% (74.5, 90.3)	41.2% (21.6, 64.0)	0.63 (0.51, 0.75)	3.66 (1.18, 11.38)	0.041[b]
Visual motor coordination	88.9% (80.2, 94.1)	35.3% (17.3, 58.7)	0.62 (0.50, 0.74)	4.36 (1.30, 14.67)	0.022[b]
Visual processing	93.8% (86.2, 97.4)	23.5% (9.6, 47.3)	0.59 (0.47, 0.71)	4.68 (1.11, 19.75)	0.046[b]
Visual attention	71.6% (61.0, 80.3)	58.8% (36.0, 78.4)	0.65 (0.53, 0.77)	3.60 (1.22, 10.61)	0.023[b]

Abbreviations: AOC, area of concern; AUC, area under the curve; CI, confidence interval; CVI, cerebral visual impairment; MON, continue to monitor; OR, odds ratio.
[a]p-values from Fisher's exact test.
[b]Statistically significant at $p < 0.05$.

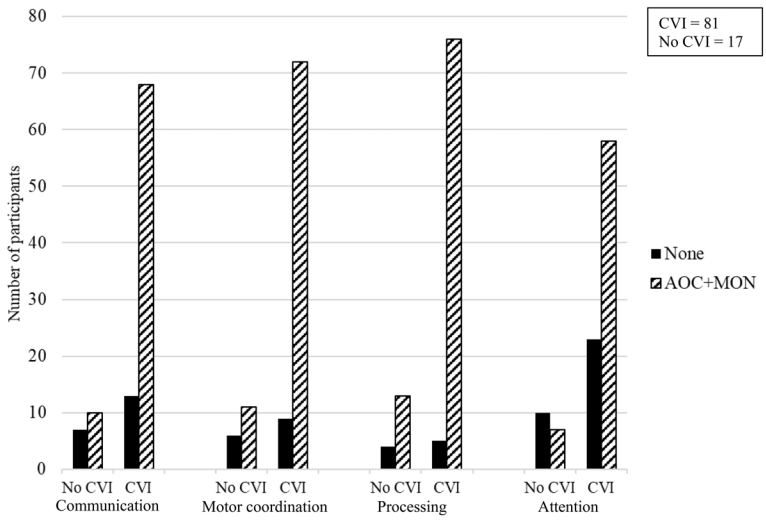


FIGURE 1 Frequency of Preverbal Visual Assessment scores by visual domain for children with and without cerebral visual impairment (CVI). Abbreviations: AOC, area of concern; MON, continue to monitor.

as AOC alone, the sensitivity and specificity of the PreViAs by domain were lower than when concern was defined as AOC + MON. Fisher's exact test was not significant for any domain of the PreViAs using AOC alone but was significant for visual communication, visual motor coordination, and visual attention when combining AOC + MON. The area under the curve values ranged from 0.52 to 0.61 for AOC alone and 0.59 to 0.65 for AOC + MON, both exhibiting wide confidence intervals. Based on sensitivity and specificity findings, concern should be defined as AOC + MON, as the model is more robust. Therefore, for the remainder of the results, AOC and MON are combined to represent the level of concern.

Figure 1 illustrates the distribution of PreViAs scores across the four visual domains, comparing children with and without CVI by level of concern (AOC + MON). The visual processing domain had the highest frequency of concern for children with CVI ($n = 76$, 94%) and without CVI ($n = 13$, 76%). Conversely, visual attention had the lowest frequency of concern for both groups, with 58 children (72%) with CVI and seven children (41%) without CVI.

The timing of the administration of the PreViAs and confirmation of CVI diagnosis varied. For children diagnosed with CVI, the PreViAs was administered before the diagnosis in 21 (26%) cases, after the diagnosis in 39 (48%) cases, and on the same day as the diagnosis in 21 (26%) cases.

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DISCUSSION

Early screening tools with high sensitivity and specificity are crucial for detecting CVI. To our knowledge, this is the first study to explore the relationship between PreViAs results and CVI diagnosis. Our group utilized retrospective data to gain a preliminary understanding of the utility of the PreViAs as a screening tool within this population. Our preliminary data suggest that the PreViAs effectively identifies children who would benefit from a CVI diagnostic evaluation, warranting further prospective studies.

As anticipated, our data showed that the PreViAs questionnaire is sensitive in detecting CVI in young children with concern. We hypothesized that AOC alone would be the most sensitive model; however, combining levels of concern (AOC + MON) demonstrated the highest sensitivity. Our results demonstrated that the PreViAs can differentiate young children with CVI from those without CVI. These findings are similar to previous studies demonstrating that the PreViAs differentiated between delayed visual maturation and typically developing individuals.[19] Our data indicates that all children screened using the PreViAs and scoring as AOC or MON in any domain should be referred for a comprehensive CVI diagnostic evaluation. Based on the results of this study, our institution has modified its clinical screening approach to refer all children who exhibit any level of concern on the PreViAs for a CVI evaluation.

Screening tools are developed to detect conditions such as CVI and other pediatric eye disorders early in life, before clinical signs become apparent or are recognized by clinicians. Screening aims to promote early detection and minimize the risk of missed diagnoses. Most screening tools maximize sensitivity to identify all individuals with a specific condition to refer them for further evaluation. High sensitivity of the PreViAs questionnaire ensures that fewer children with CVI are missed, facilitating timely interventions for optimal developmental outcomes. Early evaluations are preferable to avoid late diagnoses and missed intervention opportunities. Lower specificity of CVI screening tools can result in over-referral of individuals who are unlikely to have CVI. Since the primary aim of CVI screeners is identifying and referring children for additional testing, higher sensitivity is favored, even at the expense of specificity.

When examining the PreViAs results across all visual domains for children with CVI, our findings show that the frequency of concern and highest sensitivity were observed for the visual processing domain (94%, 93.8%), followed by visual motor coordination (89%, 88.9%), visual communication (84%, 84%), and visual attention (72%, 71.6%) respectively. A significant portion of our sample exhibit various brain abnormalities, potentially resulting in impaired visual attention.[20,21] Visual attention is an early, foundational visual skill that underlies participation in daily activities. For children with CVI, there are multiple factors that can negatively affect visual attention, including visual fatigue, sensory environment, activity familiarity, and environmental clutter.[10,22] Although we suspected that visual attention would be the domain with the highest frequency of concern, we were surprised to find that it had the lowest frequency and sensitivity; however, it demonstrated the most favorable balance between sensitivity and specificity, making it a crucial domain for early screening.

For successful, early screening of CVI, a questionnaire should be short, understandable, and easy to administer in a busy setting. The PreViAs questionnaire, with 30 simple yes/no questions, is a clinically useful tool for early CVI detection. It can be completed electronically or on paper, making it accessible for caregivers. It is feasible to use in high-risk settings, such as neonatal intensive care unit follow-up clinics, and applies to children who are developmentally aged 24 months and under who may be at risk for CVI. Studies show strong internal consistency across its visual domains, ensuring its reliability.[18,23] Alternative methods for early screening of vision impairment, primarily used in inpatient settings and requiring direct interaction with young children, may be more challenging to implement in busy, outpatient clinical settings.[24,25] Caregiver questionnaires offer several advantages over screeners that require direct administration, including convenience, time efficiency, accessibility, and cost-effectiveness. Future studies should compare the PreViAs results with direct administration vision screening tools to understand the correlations between the two.

We divided our participants into three smaller age groups to explore whether there were group differences (CVI vs non-CVI) in the mean scores of PreViAs. Formal analysis was completed for all groups (0–12 months, 13–24 months, and 25+ months). Our results showed meaningful differences between groups in all PreViAs domains in the 0- to 12-month group (CVI vs non-CVI), whereas only two domains (visual communication and visual attention) were statistically different in the 13- to 24-month group and no domains in the 25+ months group. It should be noted that the 25+ months group had a small sample size (CVI = 12 vs no CVI = 1) and statistically low power so results should be interpreted with caution. These findings align with those of Kim et al., who used the PreViAs to evaluate visual maturation in typically developing term infants 2 years of age and under to establish normative scores by domain.[26] They reported that the maturation age of certain visual perceptual skills occurred later in the second year of life. Specifically, the visual motor coordination domain matured at 24 months and visual processing at 21 months. This might explain why the visual motor coordination and visual processing domains were not statistically different in our study's 13- to 24-month group. The visual perceptual skills that Kim et al. reported maturing in the first year of life were visual attention at 6 months and visual communication at 12 months.[26] These data support our results in that visual attention and visual communication were significant in both the 0- to 12-month and 13- to 24-month groups. It may be easier for caregivers to notice delays in visual attention and visual communication during the first 12 months since these domains mature earlier. Since visual motor coordination and visual processing mature later and visual delays may be masked by the development

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of mobility and language skills, deficits in these areas may be harder for caregivers to detect. Another study used magnetic resonance imaging data collected at term-adjusted age to show that white matter tracts in infants born preterm were associated with PreViAs scores collected between 3- and 4-months corrected age.[27] These findings support our results that early detection of CVI is possible within the first year of life. The clinical implication of our findings in combination with previous studies support that the PreViAs screener may be most accurate in detecting CVI in the first year of life, especially focusing on the domains of visual attention and visual communication. However, more data are needed utilizing a larger, prospective cohort to verify these findings.

It is important to note in this study the PreViAs was administered as part of a CVI evaluation in children who were referred because of concern for CVI or who had an existing diagnosis of CVI. As such, the PreViAs was used to characterize visual development rather than as a standalone screening tool. It was typically completed within 1 to 6 months after a CVI diagnosis or as part of the CVI diagnostic evaluation. However, as shown in our data (Table 1), many children in our study had identifiable conditions, putting them at higher risk for CVI early in life. This suggests that the PreViAs could be tested prospectively as a true CVI screening tool in future studies for children with high-risk conditions. Caregivers of children with diagnoses known to increase the risk for CVI could complete the PreViAs early in life to ascertain if more comprehensive diagnostic CVI testing is warranted. This approach would allow for a more accurate determination of the sensitivity and specificity of the PreViAs within a broader, high-risk population. Interestingly, the PreViAs was completed for two children with minimal risk for CVI. These children were ultimately diagnosed with anterior visual pathway disease and not brain-based visual impairment. Of note, the PreViAs appropriately screened these children as negative for CVI.

The PreViAs screening questionnaire demonstrated clinical usefulness in our study. Limitations of this study include the retrospective nature of this study and small sample size at a single institution. Our data exhibited wide 95% confidence intervals for several odds ratios which likely reflect the small sample size and limited number of non-CVI cases (especially in the 25+ months group). Our findings are preliminary and should be confirmed in larger, higher-powered studies. Additionally, this study was exploratory in nature with the primary objective to identify potential associations; therefore, we did not apply formal multiple testing adjustments (e.g. Bonferroni, Holm) to control for the family-wise error rate. Therefore, our findings should be interpreted with caution as they may have increased risk of type one error inherent in exploratory analysis and should be confirmed in larger, prospective studies.

Our sample was derived from children who were referred for a CVI evaluation; therefore our sample was skewed toward those who received a diagnosis of CVI. Additionally, the PreViAs was completed as a part of the CVI evaluation, not as a true screener. Future studies should recruit young children with co-occurring medical conditions commonly associated with CVI to capture a broader, high-risk population to evaluate the PreViAs as a true screening tool. This approach would allow for a better understanding of the sensitivity and specificity of the PreViAs in a wider, high-risk demographic. Future prospective studies should also evaluate the PreViAs as a function of age and CVI severity. Despite these limitations, the findings from this study can be generalized to young children with comorbid medical diagnosis that put them at higher risk for CVI.

CONCLUSIONS

This study is the first to evaluate the PreViAs screening questionnaire among children with suspected CVI. Our results provide preliminary evidence that the PreViAs is a sensitive screening tool for detecting CVI early in life. Our data shows that children who score in the AOC or MON categories on any visual domain of the PreViAs should be monitored closely and referred for a comprehensive CVI evaluation for early diagnosis.

Although this study is limited by its retrospective design and small sample size skewed toward CVI diagnoses, it is the first study to explore the relationship between the PreViAs and CVI diagnosis. Future studies should prospectively utilize the PreViAs as a screening tool in a broader, high-risk group of children with medical conditions commonly associated with CVI. Future prospective studies should also examine the correlation between the PreViAs and other measures of vision function and functional vision.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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